TP – IAM - KEYCLOAK

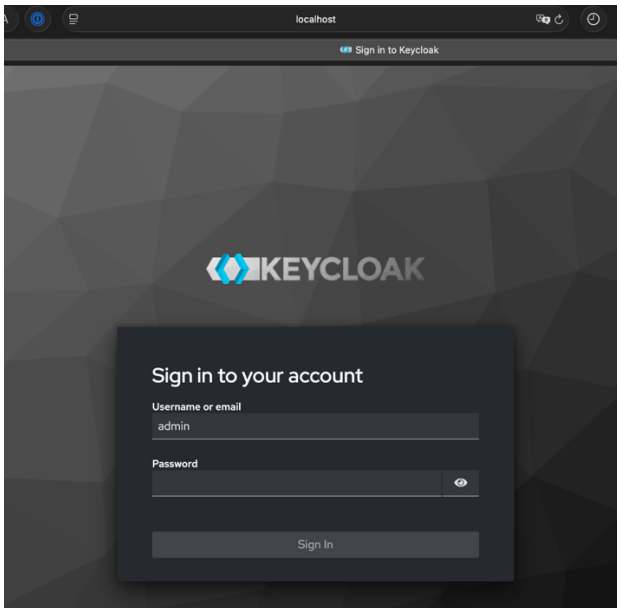
Ludivine PERRIER BABIN - 20250108

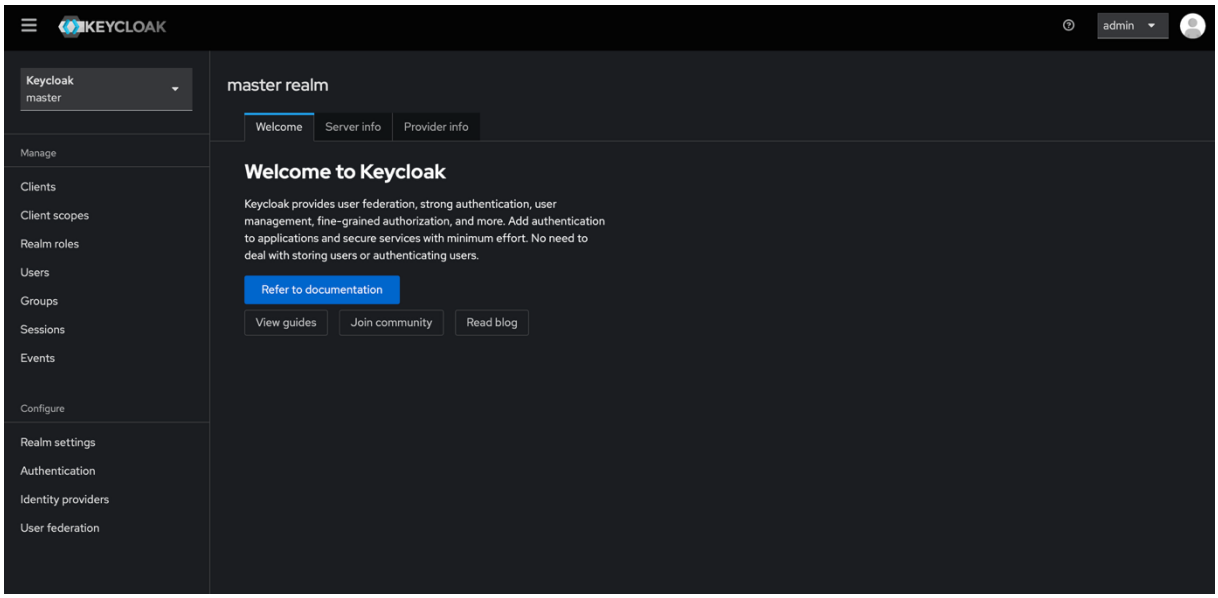
EFREI – M1 CSM – 2025-2026

Création du docker :

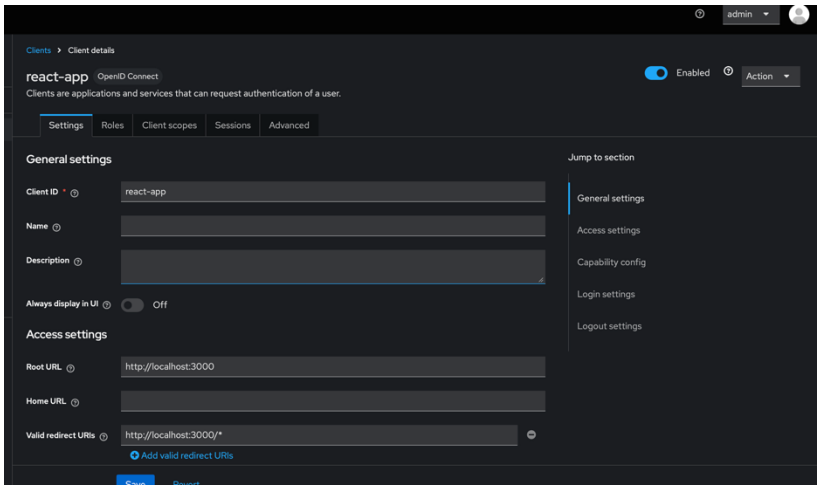
```
lud docker run -p 8080:8080 \
-e KEYCLOAK_ADMIN=admin \
-e KEYCLOAK_ADMIN_PASSWORD=admin1 \
quay.io/keycloak/keycloak:26.0.1 start-dev
Updating the configuration and installing your custom providers, if any. Please wait.
2026-04-28 14:20:40,530 INFO  [io.qua.dep.QuarkusAugmentor] (main) Quarkus augmentation completed in 7502ms
2026-04-28 14:20:43,865 INFO  [org.keycloak.quarkus.runtime.storage.infinispan.CacheManagerFactory] (main) Starting Inf
inispan embedded cache manager
2026-04-28 14:20:44,009 INFO  [org.keycloak.quarkus.runtime.storage.infinispan.CacheManagerFactory] (main) Persistent u
ser sessions enabled and no memory limit found in configuration. Setting max entries for sessions to 10000 entries.
2026-04-28 14:20:44,009 INFO  [org.keycloak.quarkus.runtime.storage.infinispan.CacheManagerFactory] (main) Persistent u
ser sessions enabled and no memory limit found in configuration. Setting max entries for clientSessions to 10000 entrie
s.
2026-04-28 14:20:44,009 INFO  [org.keycloak.quarkus.runtime.storage.infinispan.CacheManagerFactory] (main) Persistent u
ser sessions enabled and no memory limit found in configuration. Setting max entries for offlineSessions to 10000 entri
es.
2026-04-28 14:20:44,009 INFO  [org.keycloak.quarkus.runtime.storage.infinispan.CacheManagerFactory] (main) Persistent u
ser sessions enabled and no memory limit found in configuration. Setting max entries for offlineClientSessions to 10000
 entries.
2026-04-28 14:20:44,317 INFO  [org.infinispan.CONTAINER] (ForkJoinPool.commonPool-worker-1) ISPN000556: Starting user m
arshaller 'org.infinispan.commons.marshall.ImmutableProtoStreamMarshaller'
2026-04-28 14:20:45,792 INFO  [org.keycloak.quarkus.runtime.storage.database.liquibase.QuarkusJpaUpdaterProvider] (main
) Initializing database schema. Using changelog META-INF/jpa-changelog-master.xml

UPDATE SUMMARY
Run:                          144
Previously run:                 0
Filtered out:                   0
-------------------------------
Total change sets:            144
```

Page d'accueil de Keycloak :

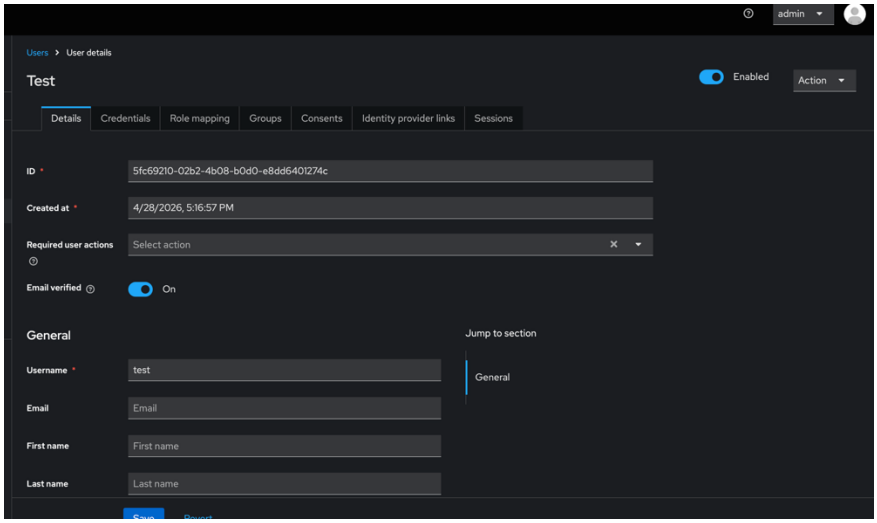




Création du client pour l'application :



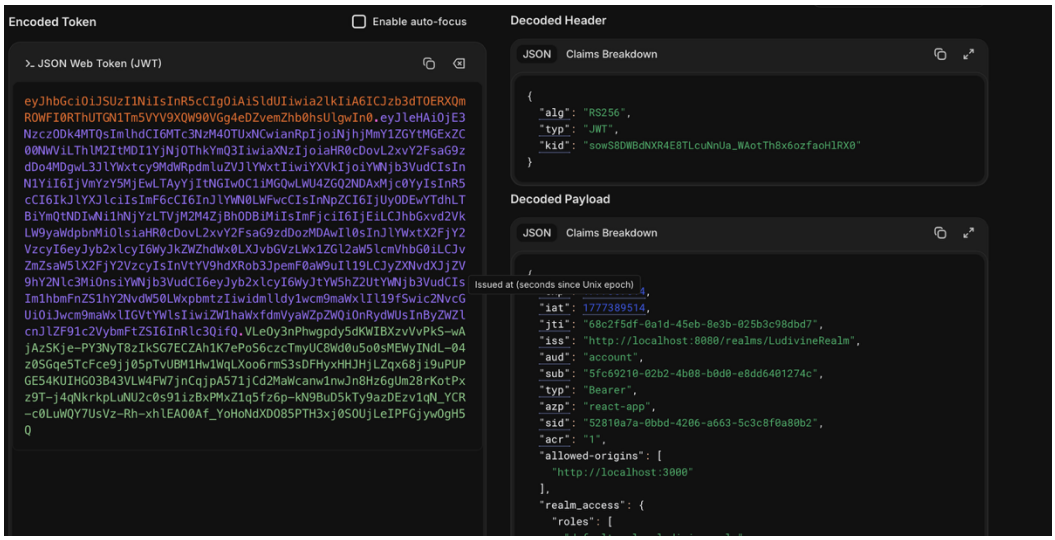
Création du compte user pour la connexion :



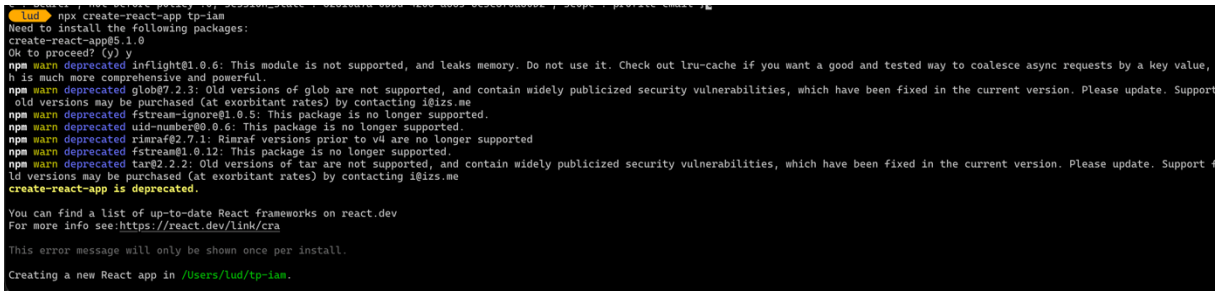
Vérification du token :

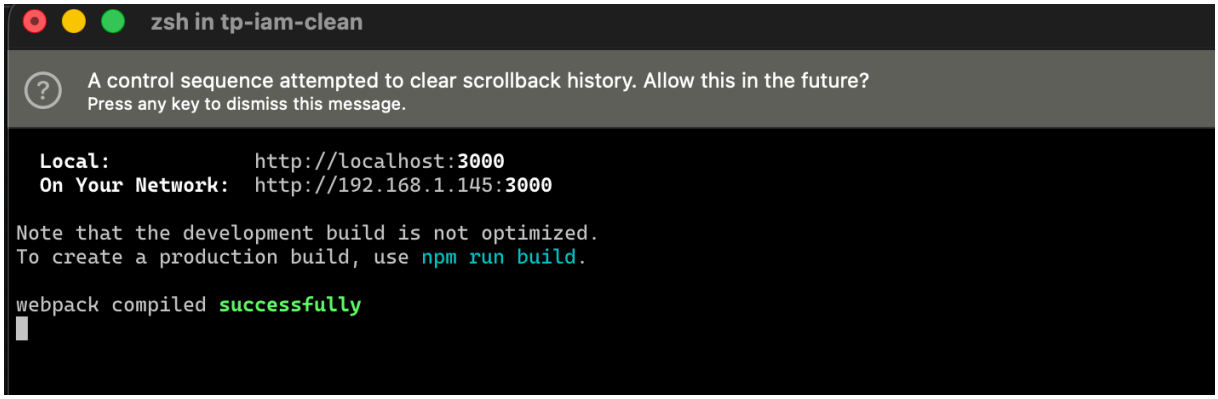
```
lud curl -X POST "http://localhost:8080/realms/LudivineRealm/protocol/openid-connect/token" \
-H "Content-Type: application/x-www-form-urlencoded" \
-d "client_id=react-app" \
-d "username=test" \
-d "password=test" \
-d "grant_type=password"
{"access_token":"eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJzb3dTOERXQmROWFI0RThUTGN1Tm5VYV9XQW90VGg4eDZvemZhb0hsUlgwIn0.eyJleHAiOjE3NzczODk4MTQsImlhdCI6MTc3NzM4OTUxNCwianRpIjoiNjhjMmY1ZGYtMGE
xZC00NWViLThlM2ItMDI1YjNjOThkYmQ3IiwiaXNzIjoiaHR0cDovL2xvY2FsaG9zdDo4MDgwL3JlYWxtcy9MdWRpdmluZVJlYWxtIiwiYXVkIjoiYWNjb3VudCIsInN1YiI6IjVmYzY5MjEwLTAyYjItNGIwOCliMGQwLWU4ZGQ2NDAxMjc0YyIsInR5cCI6IkJlYXJ
lciIsImF6cCI6InJlYWN0LWFwcCIsInNpZCI6IjUyODEwYTdhLTBiWQtNDIwNi1hNjYzLTVjM2M4Zj8hODBiMiIsImFjciI6IjEiLCJhbGxvd2VkLW9yaWdpbnMiOlsiaHR0cDovL2xvY2FsaG9zdDozMDAwIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJkZWZ
hdWxBLXJybGVzLWx1ZGl2aWBlcmVhbG01LCJvZmZsaW51X2FjY2VzcyIsInVtYV9hdXRob3JpemFBaW9uIl19LCJyZXNvdXJjZV9hY2Nlc3MiOnsiYWNjb3VudCI6eyJyb2xlcyI6WyJtYW5hZ2UtYWNjb3VudCIsImlhbmFnZSIhY2NvdW50LWxpbmtzIiwidmlldyI
wcm9maWxlIl19fSwic2NvcGUiOiJwcm9maWxlIGVtYWlsIiwiZW1haWxfdmVyaWZpZWQiOnRydWUsInByZWZlcnJlZF91c2VybmFtZSI6InRlc3QifQ.VLeOy3nPhwgpdy5dKWIBXzvVvPk5-wAjAzSkje-PY3NyT8zIkSG7ECZAh1K7ePoS6czcTmyUC8Wd0u5o0sME
WyINdL-O4z0SGqe5TcFce9jj05pTvUBM1Hw1WqLXoo6rmS3sDFHyxHHJHjLZqx68ji9uPUPGE54KUIHGO3B43VLW4FW7jnCqjpA571jCd2MaWcanw1nwJn8Hz6gUm28rKotPxz9T-j4qNkrkpLuNU2c0s91izBxPMxZ1q5fz6p-kN9BuD5kTy9azDEzv1qN_YCR-c0Lu
WQY7UsVz-Rh-xhlEAO0Af_YoHoNdXDO85PTH3xj0SOUjLeIPFGjywOgH5Q","expires_in":300,"refresh_expires_in":1800,"refresh_token":"eyJhbGciOiJIUzUxMiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJiNjQ5ZWEyZi1mMTlkLTQ0MGYtODFm
MC02YzkyMGFiM2MzOGQifQ.eyJleHAiOjE3NzczOTEzMTQsImlhdCI6MTc3NzM4OTUxNCwianRpIjoiMTdiMDM2MjctZmVlYy00YzU5LWE5ZTYtMWM0ZDkxZDdiZDg0IiwiaXNzIjoiaHR0cDovL2xvY2FsaG9zdDo4MDgwL3JlYWxtcy9MdWRpdmluZVJlYWxtIiwiY
XVkIjoiaHR0cDovL2xvY2FsaG9zdDo4MDgwL3JlYWxtcy9MdWRpdmluZVJlYWxtIiwic3viIjoiNWZjNjkyMTAtMDJiMi00YjA4LWIwZDAtZThkZDY0MDEyNzRjIiwidHlwIjoiUmVmcmVzaCIsImF6cCI6InJlYWN0LWFwcCIsInNpZCI6IjUyODEwYTdhLTBiYmQtN
DIwNi1hNjYzLTVjM2M4Zj8hODBiMiIsInNjb3BlIjoiYWNyIHdlYi1vcmlnaW5zIGJhc2ljIHByb2ZpbGUgcm9sZXMgZW1haWwifQ.mF5u8dk5PI8ypSICAGiPzBSRKTcgCh5VZaPo49vmln1K7c2ZAWiWqmVYiqwyX6rLTKy86q_RxZ-3oZv-_wtPHA","token_typ
e":"Bearer","not-before-policy":0,"session_state":"52810a7a-0bbd-4206-a663-5c3c8f0a80b2","scope":"profile email"}%
```

Décryptage du token sur jwt.io :



Création et lancement de l'application react :





Test de connexion avec le compte test :

Page privée 🔒

exp	1777392176
iat	1777391876
jti	"948c2950-9f5f-430b-9211-e303fbc7fdb6"
iss	"http://localhost:8080/realms/LudivineRealm"
aud	"account"
sub	"5fc69210-02b2-4b08-b0d0-e8dd6401274c"
typ	"Bearer"
azp	"react-app"
sid	"20f77d53-b6e4-4786-b121-5e0d716d485f"
acr	"1"
allowed-origins	["http://localhost:3000"]
realm_access	{"roles":["default-roles-ludivinerealm","offline_access","uma_authorization"]}
resource_access	{"account":{"roles":["manage-account","manage-account-links","view-profile"]}}
scope	"profile email"
email_verified	true
preferred_username	"test"

Lien repo Git :

https://github.com/LudOhm/TP-IAM-keycloak